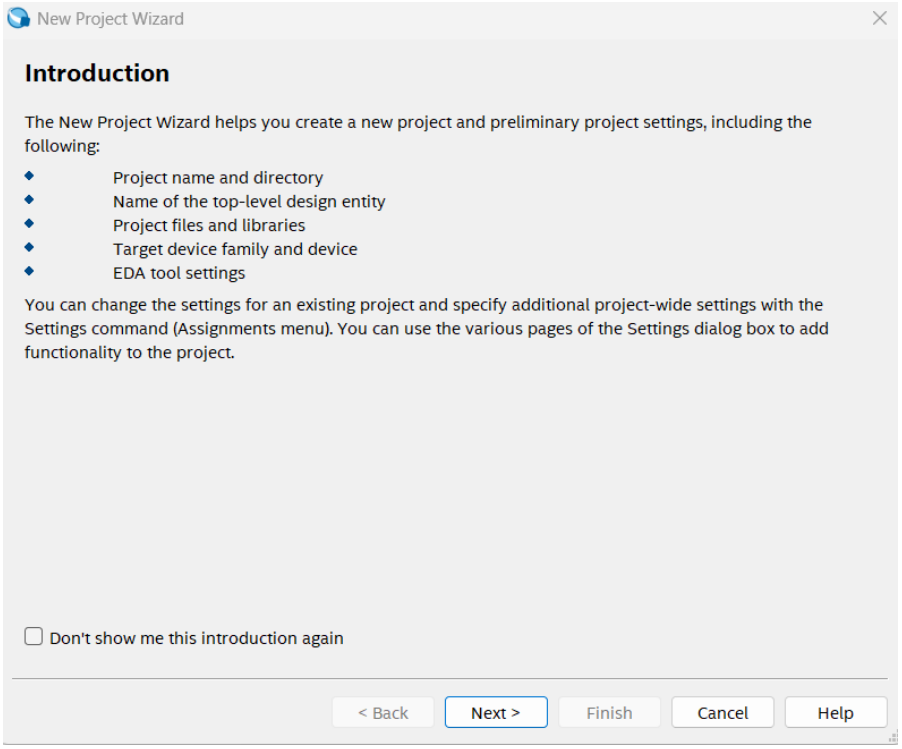
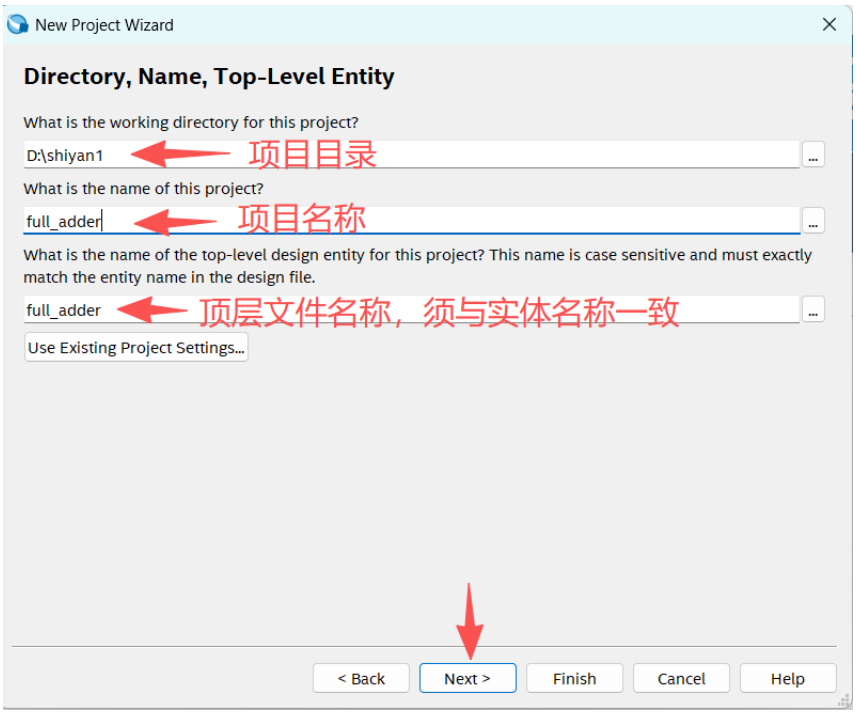
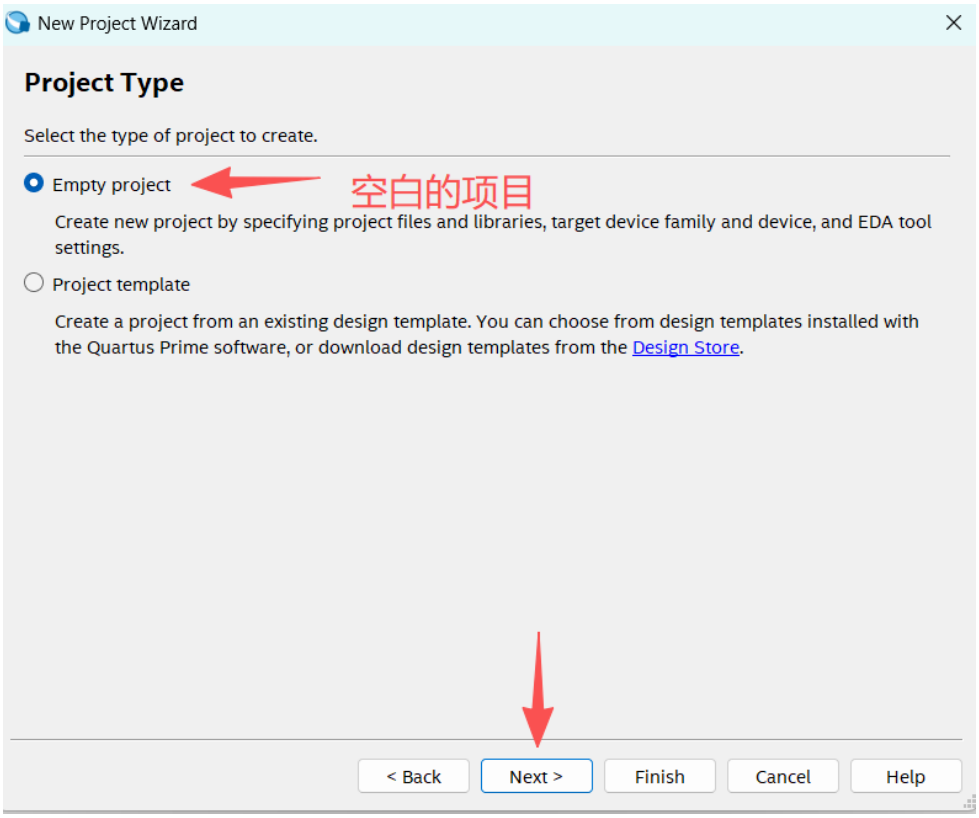
Quartus Prime 18.0　Lite Edition　基础操作示例

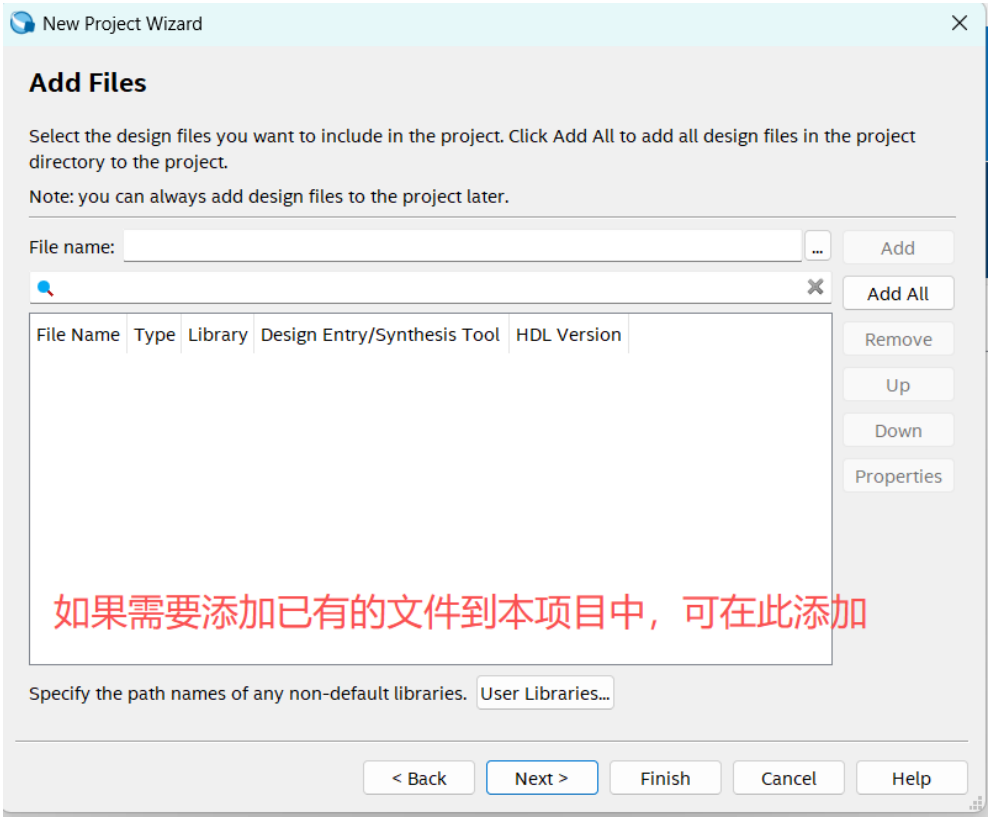
1.新建工程

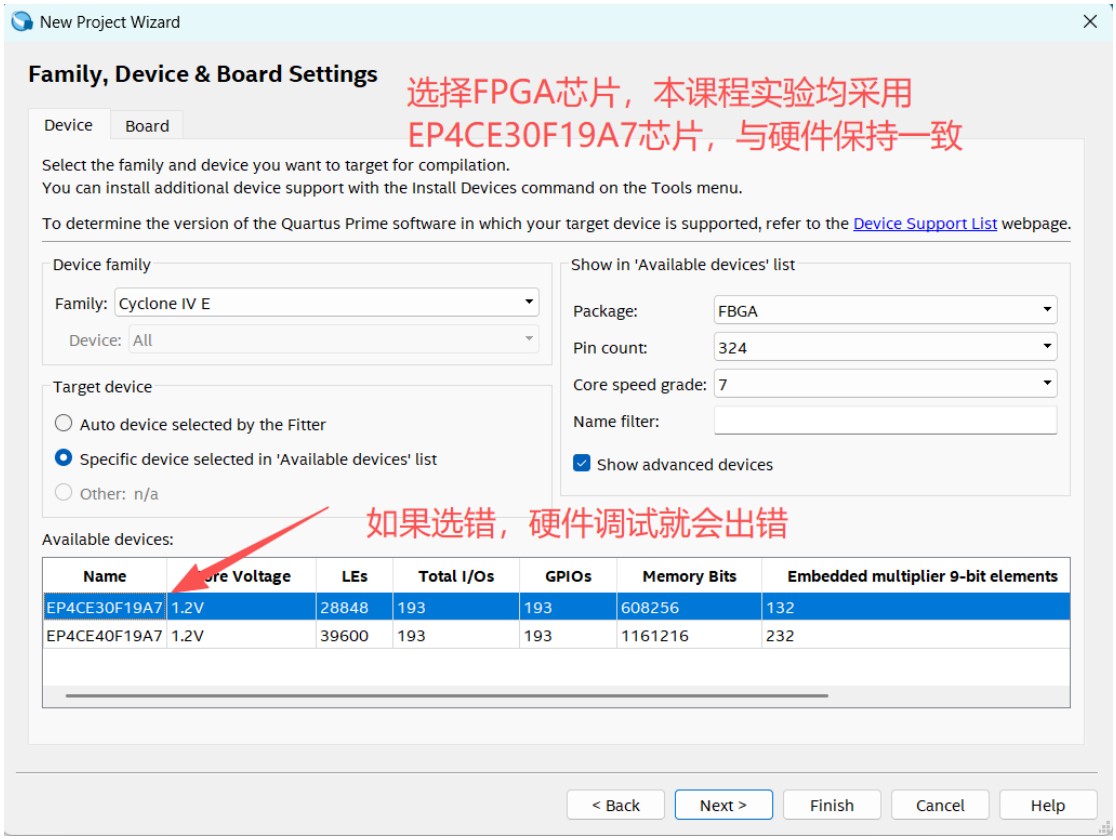
（1）File → New Project Wizard

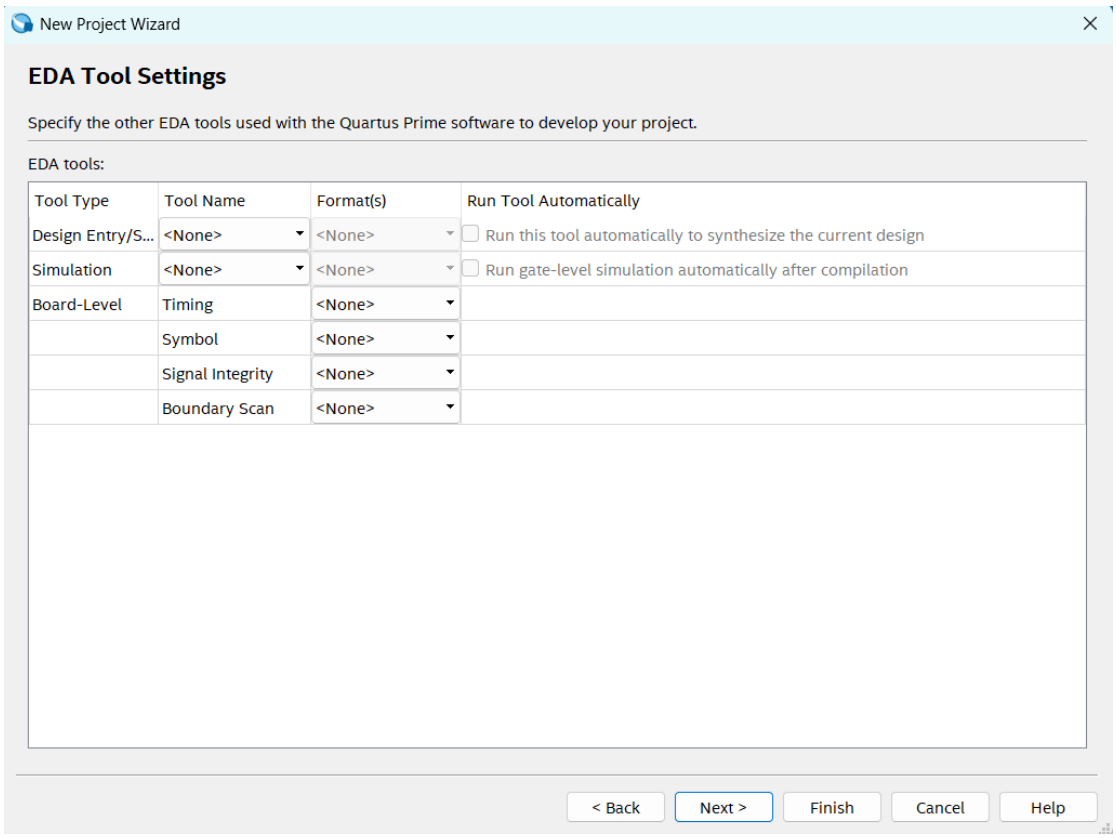


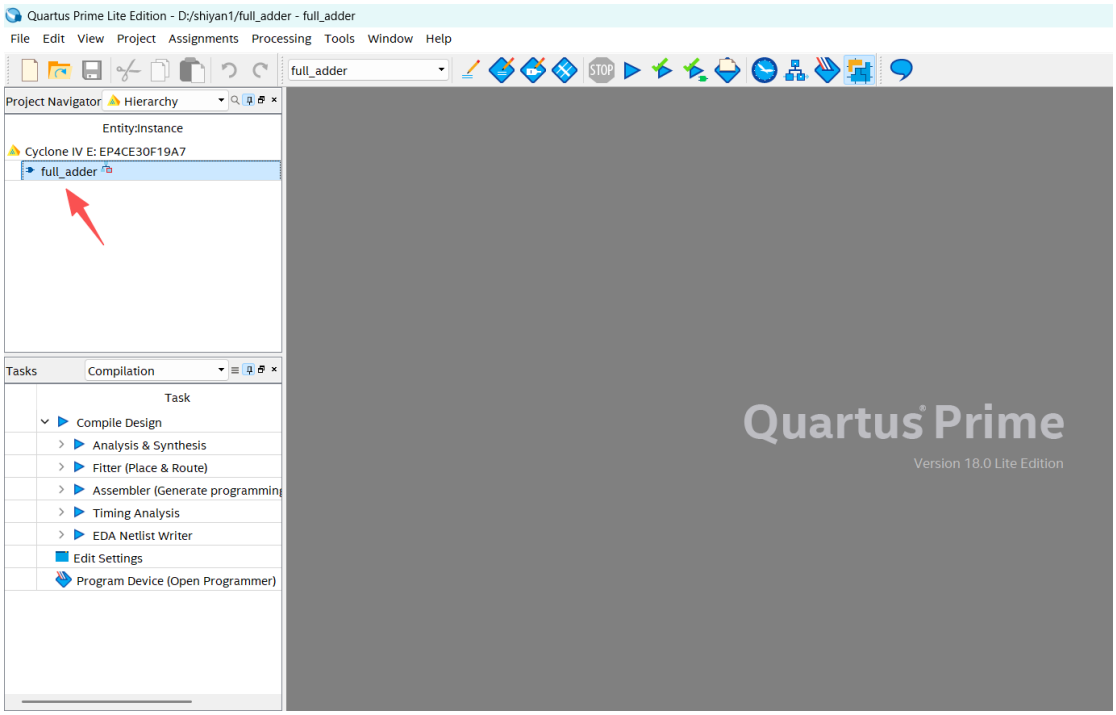



New Project Wizard
Project Type
Select the type of project to create.
Empty project
空白的项目
Create new project by specifying project files and libraries, target device family and device, and EDA tool settings.
Project template
Create a project from an existing design template. You can choose from design templates installed with the Quartus Prime software, or download design templates from the Design Store.
< Back
Next >
Finish
Cancel
Help


New Project Wizard
Add Files
Select the design files you want to include in the project. Click Add All to add all design files in the project directory to the project.
Note: you can always add design files to the project later.
File name:
Add
Add All
Remove
Up
Down
Properties
File Name
Type
Library
Design Entry/Synthesis Tool
HDL Version
如果需要添加已有的文件到本项目中，可在此添加
Specify the path names of any non-default libraries.
User Libraries...
< Back
Next >
Finish
Cancel
Help


New Project Wizard
Family, Device & Board Settings
选择FPGA芯片，本课程实验均采用
EP4CE30F19A7芯片，与硬件保持一致
Device
Board
Select the family and device you want to target for compilation.
You can install additional device support with the Install Devices command on the Tools menu.
To determine the version of the Quartus Prime software in which your target device is supported, refer to the Device Support List webpage.
Device family
Family: Cyclone IV E
Device: All
Target device
Auto device selected by the Fitter
Specific device selected in 'Available devices' list
Other: n/a
Show in 'Available devices' list
Package: FBGA
Pin count: 324
Core speed grade: 7
Name filter:
Show advanced devices
如果选错，硬件调试就会出错
Available devices:
Name
re Voltage
LEs
Total I/Os
GPIOs
Memory Bits
Embedded multiplier 9-bit elements
EP4CE30F19A7 1.2V 28848 193 193 608256 132
EP4CE40F19A7 1.2V 39600 193 193 1161216 232
< Back
Next >
Finish
Cancel
Help

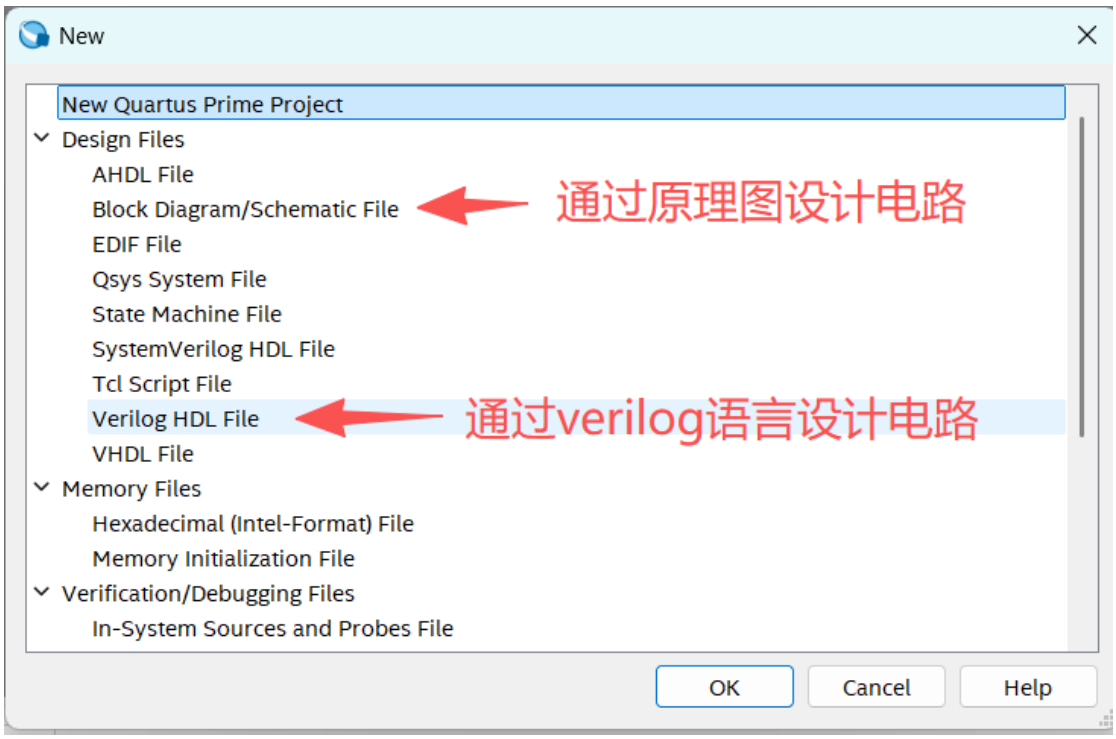

New Project Wizard
EDA Tool Settings
Specify the other EDA tools used with the Quartus Prime software to develop your project.
EDA tools:
Tool Type
Tool Name
Format(s)
Run Tool Automatically
Design Entry/S... <None> <None> Run this tool automatically to synthesize the current design
Simulation <None> <None> Run gate-level simulation automatically after compilation
Board-Level Timing <None>
Symbol <None>
Signal Integrity <None>
Boundary Scan <None>
< Back
Next >
Finish
Cancel
Help

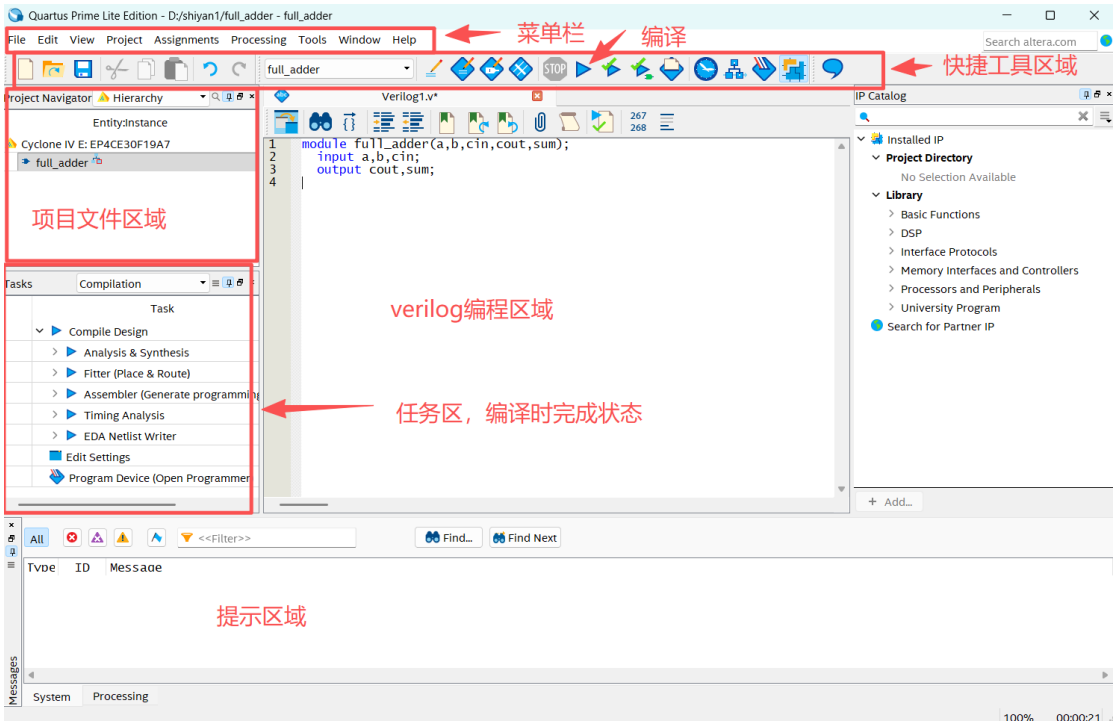


2.创建文件

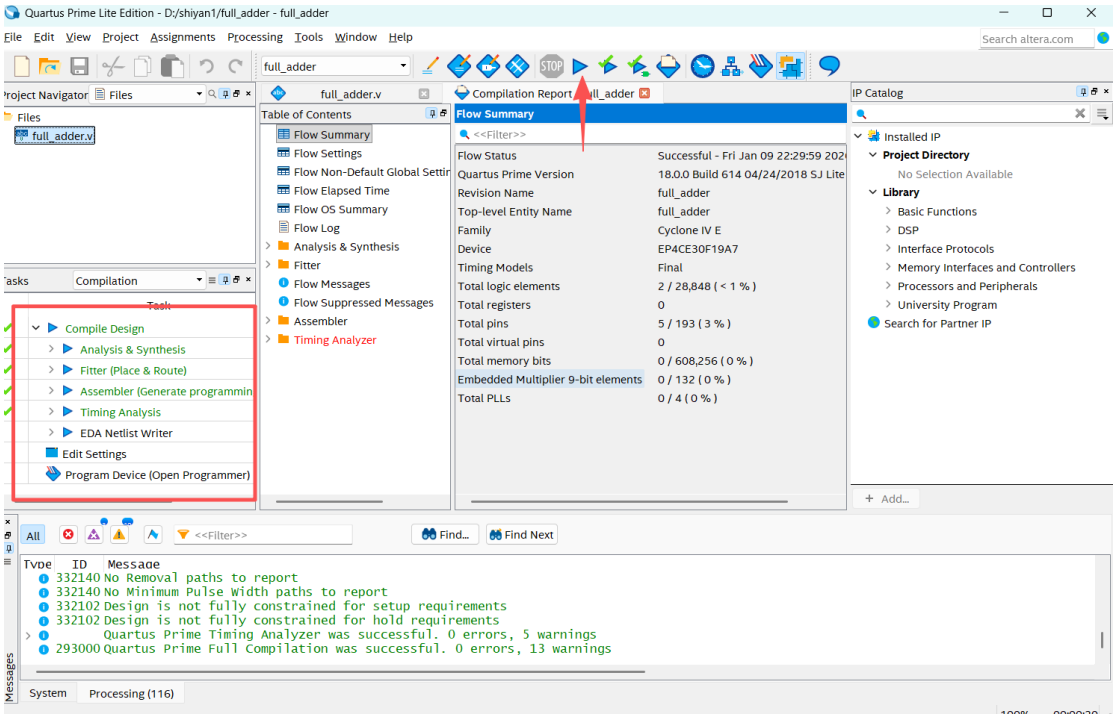
File-New

创建 verilog 元件





3.编译



4.生成自定义元件符号

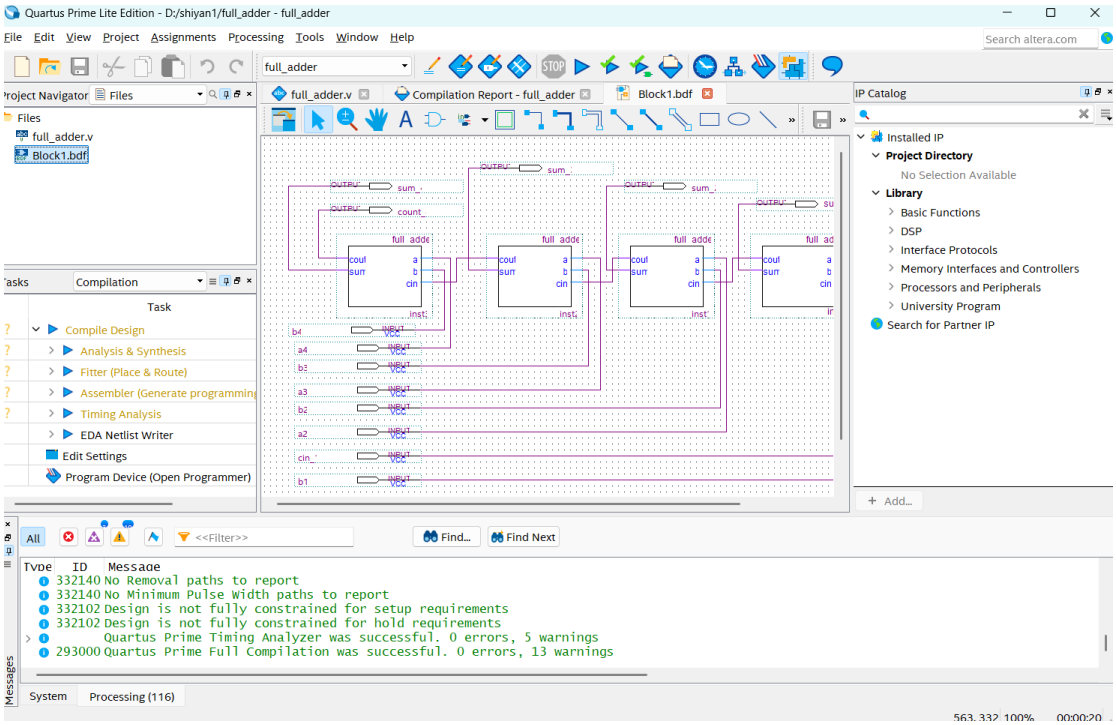
生成自定义元件符号（Hierarchy 下拉菜单选择 files，可以看到所有文件，鼠标右键在文件上面 Create Symbol Files for Current File

可以创建成元器件)

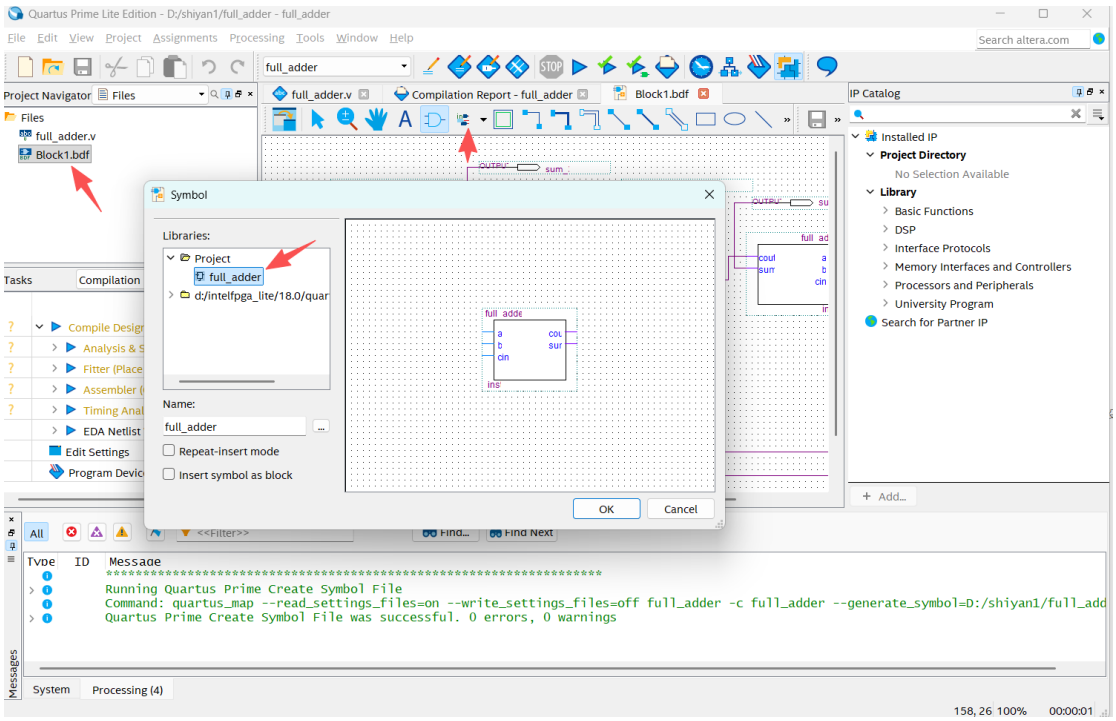
File → Create/Update → Create Symbol Files for Current File

5.原理图设计方法，创建原理图（四位加法器，相当于第二步）

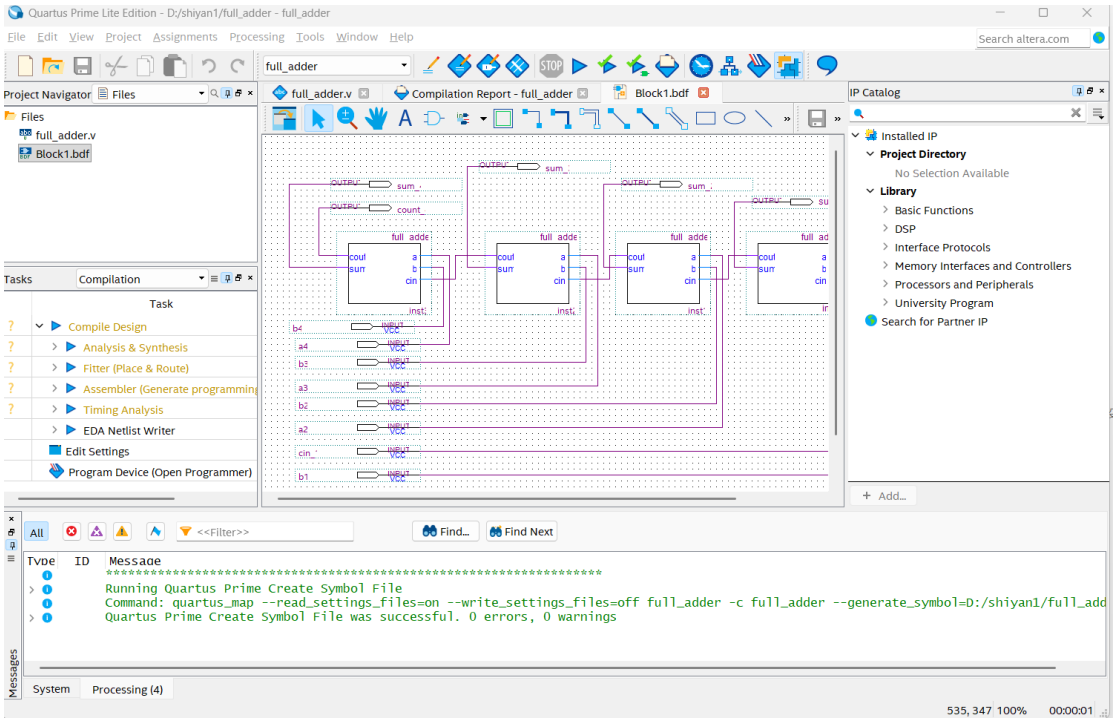
创建 NEW—Block Diagram



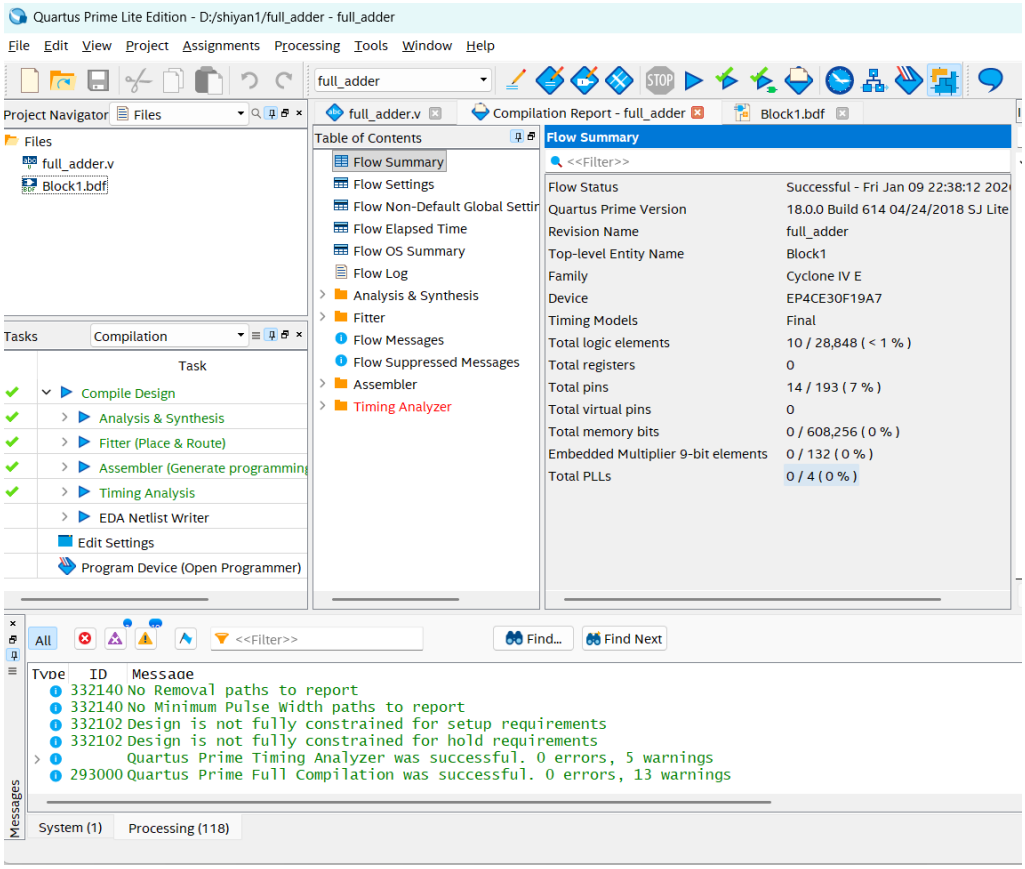
设计(鼠标双击设计区域，会弹出元器件对话框，我们自己生成的元件在 project 里面，还有输入输出在元器件库的 primitives-pin（input，output 里面)，连线使用鼠标，当鼠标靠近连接点时会变成十字)



设计(鼠标双击设计区域)

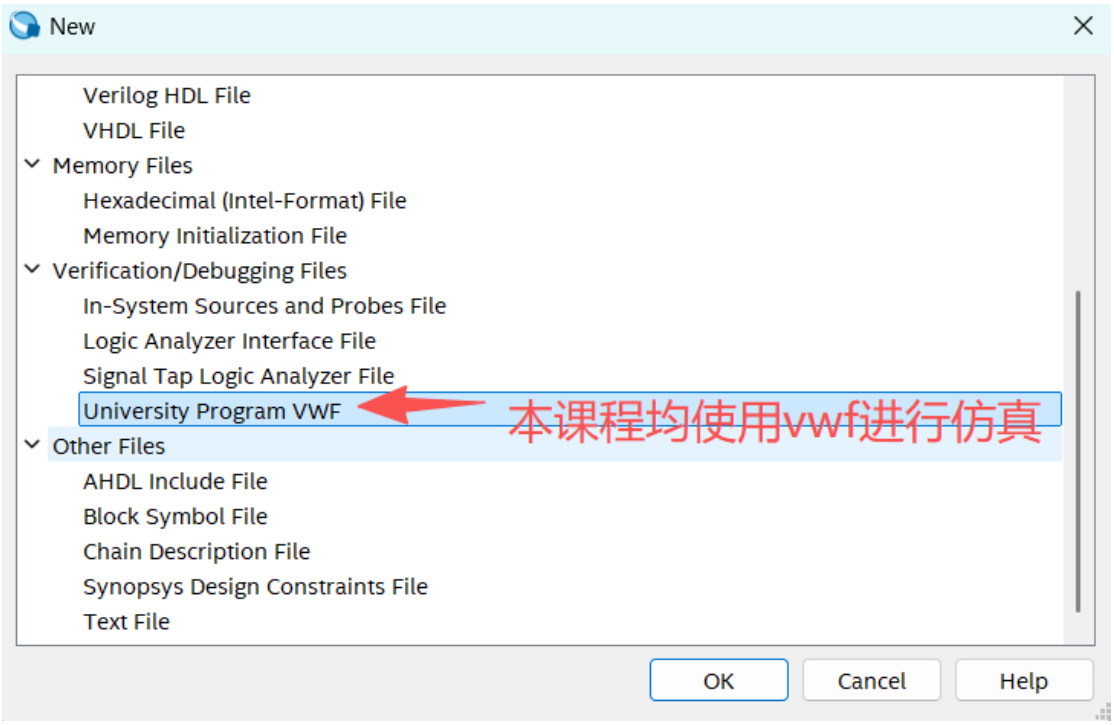


编译，在编译前先设置原理图文件为顶层实体文件，相当于项目的入口，非常重要。在 files 里面的文件上面点击右键，选择 set as top-level entity,创建顶层文件。

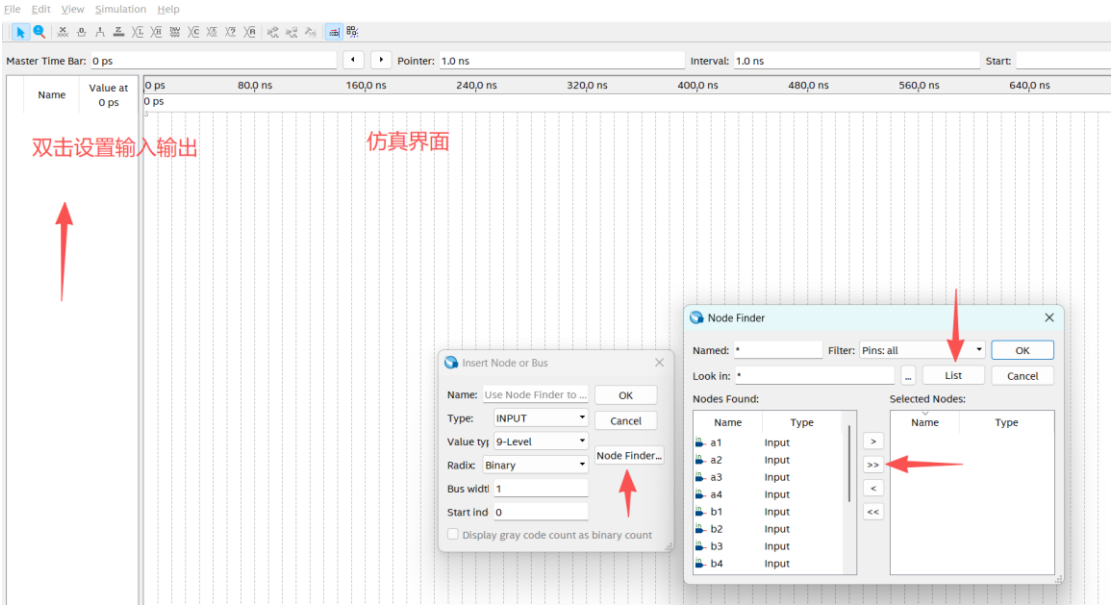


6.仿真

创建仿真文件，本课程均使用 vwf 进行波形仿真，File-NEW



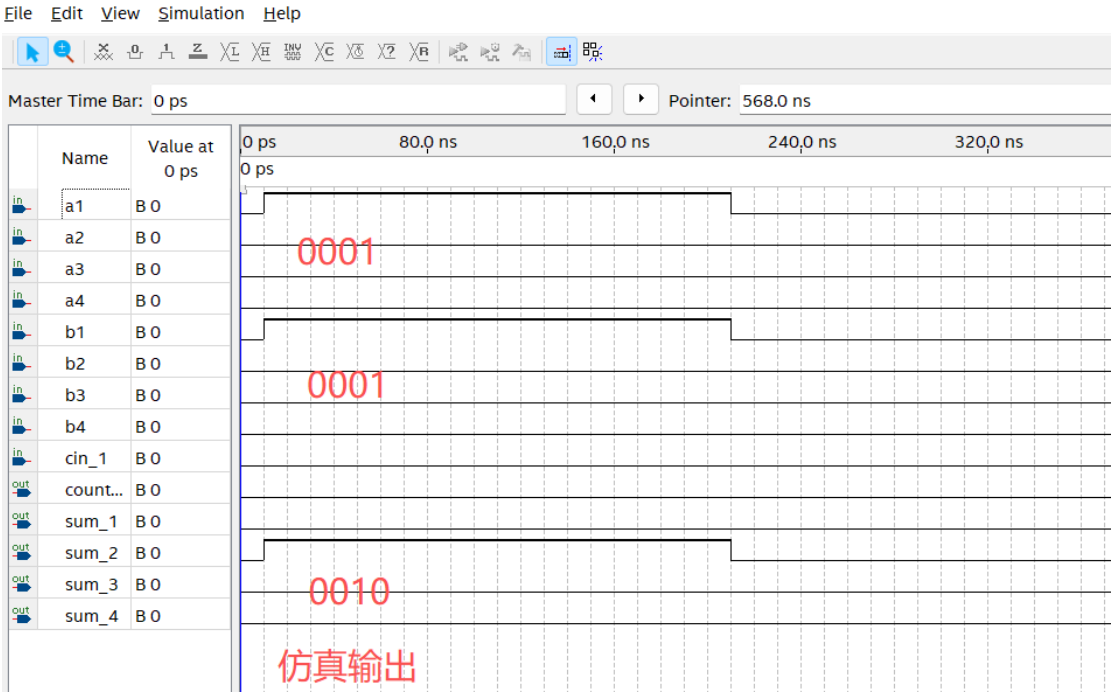
设置输入输出



设置输入值，运行仿真查看结果

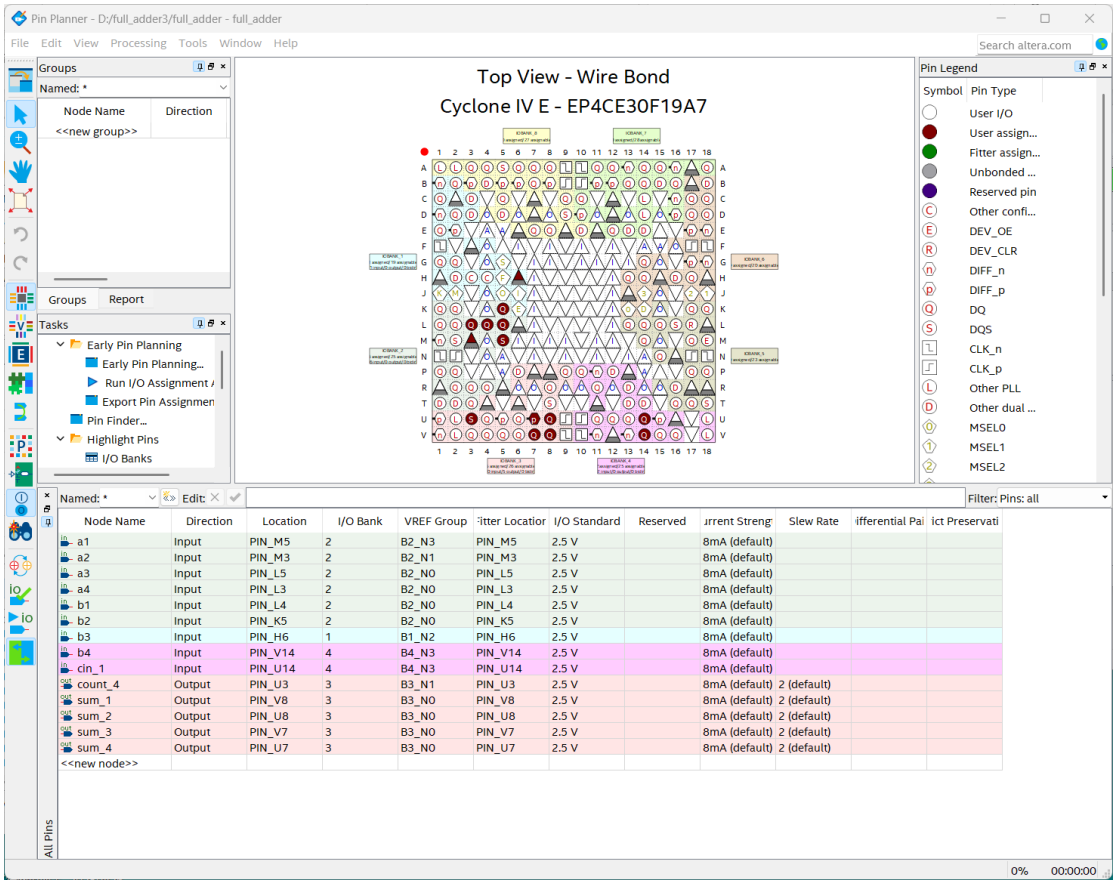
输入设置：鼠标按住输入的波形区域拖动选择，然后可以选择高低电平。输入高电平代表 1，低电平代表 0，输入完成后，点击运行（run

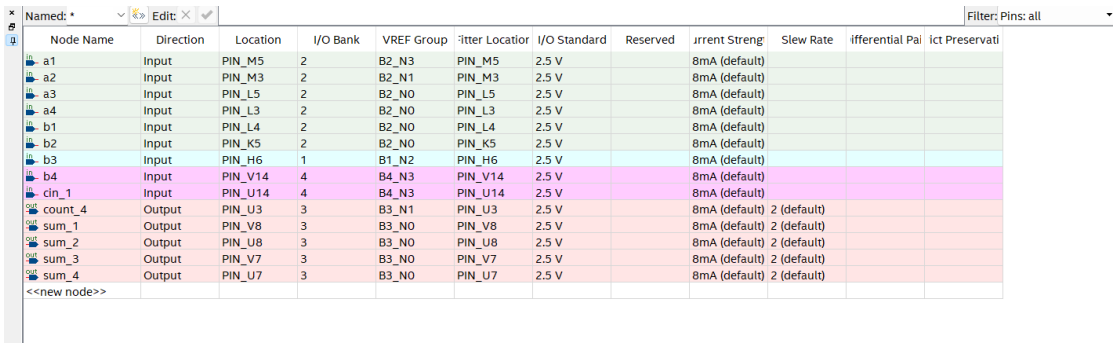
FUNCTIONAL 查看仿真结果)



7.引脚绑定

Assignments-pin planner 进入引脚绑定选项卡





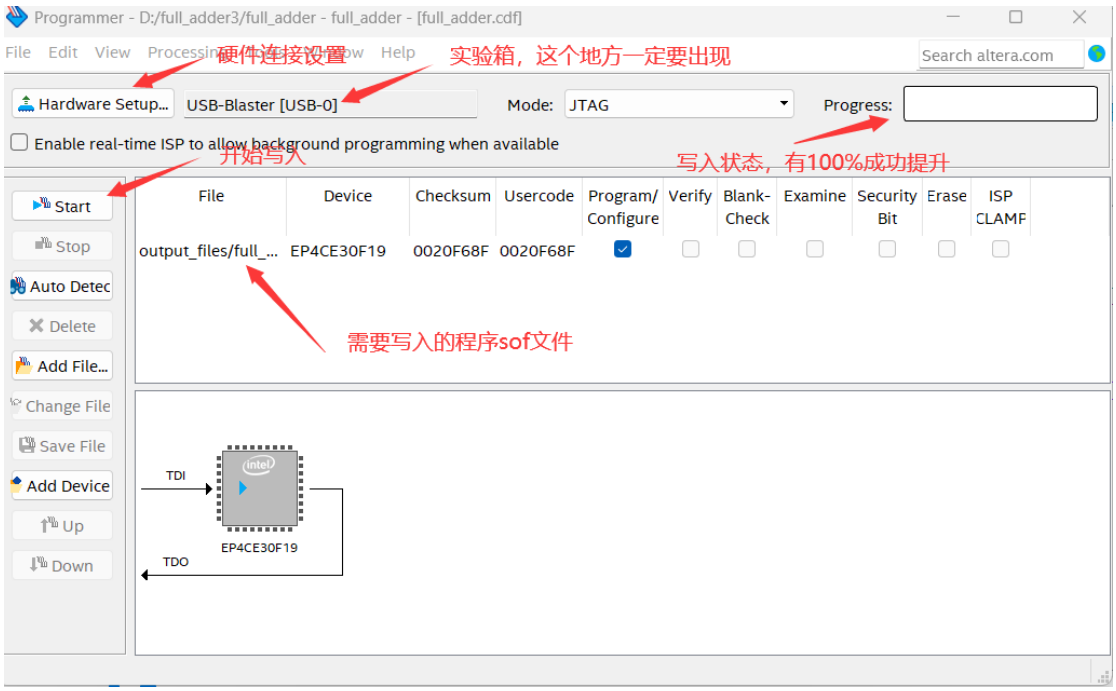
这个引脚与实验指导书后面对应的引脚设置 Location(此处注意规划好输入输出，实验一采用波动开关以及 led 灯作为输入输出即可，后面的实验还需使用按键、键盘、数码管作为输入输出，可以查询相关资料)

引脚绑定后一定要重新编译。

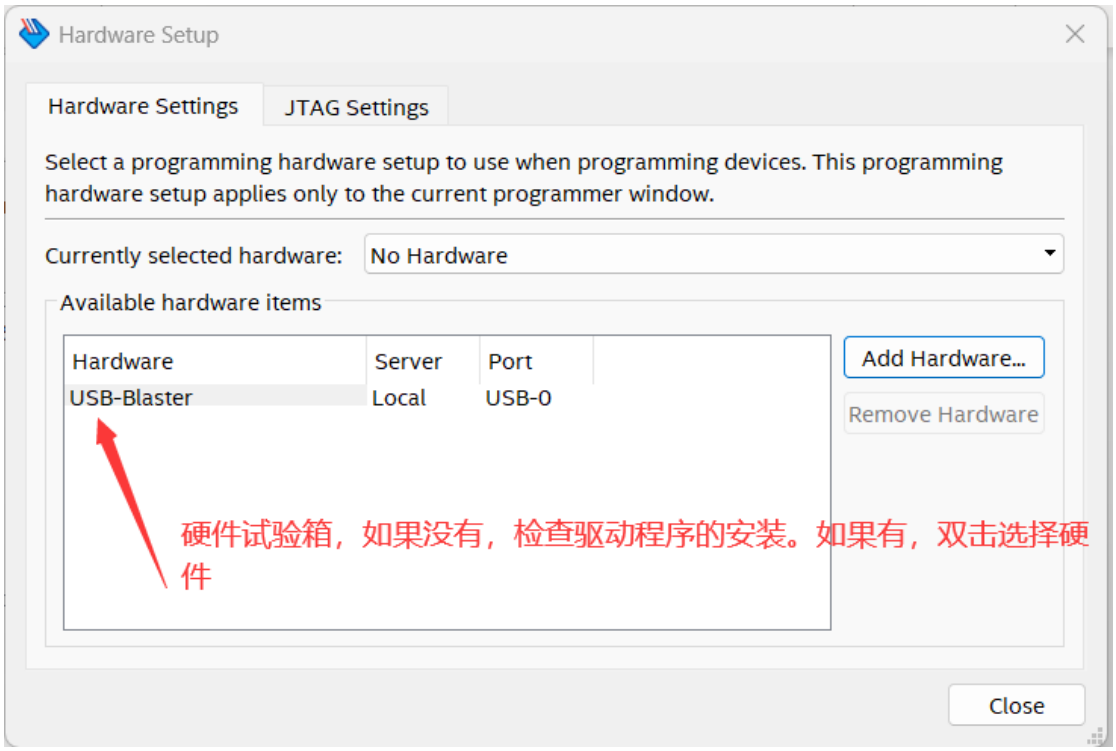
8.硬件试验箱验证

（1）连接试验箱

Toos—Programmer 进入



实验箱设置



驱动程序安装：

查看硬件是否连接：hardware setup 如果没有发现硬件，检查驱动程序是否正常，此电脑右键管理--设备管理器 --有没有 usb-blaster 硬件是黄色感叹号的，如果有，点击右键更新程序--选择本地计算机--浏览 quartus 安装目录（根据实际情况：D:\intelFPGA_lite\18.0\quartus\drivers）安装更新即可。

硬件设置好后，点击 start，开始写入硬件，直到进度为 100%，然后就可以在实验箱上面操作验证。